

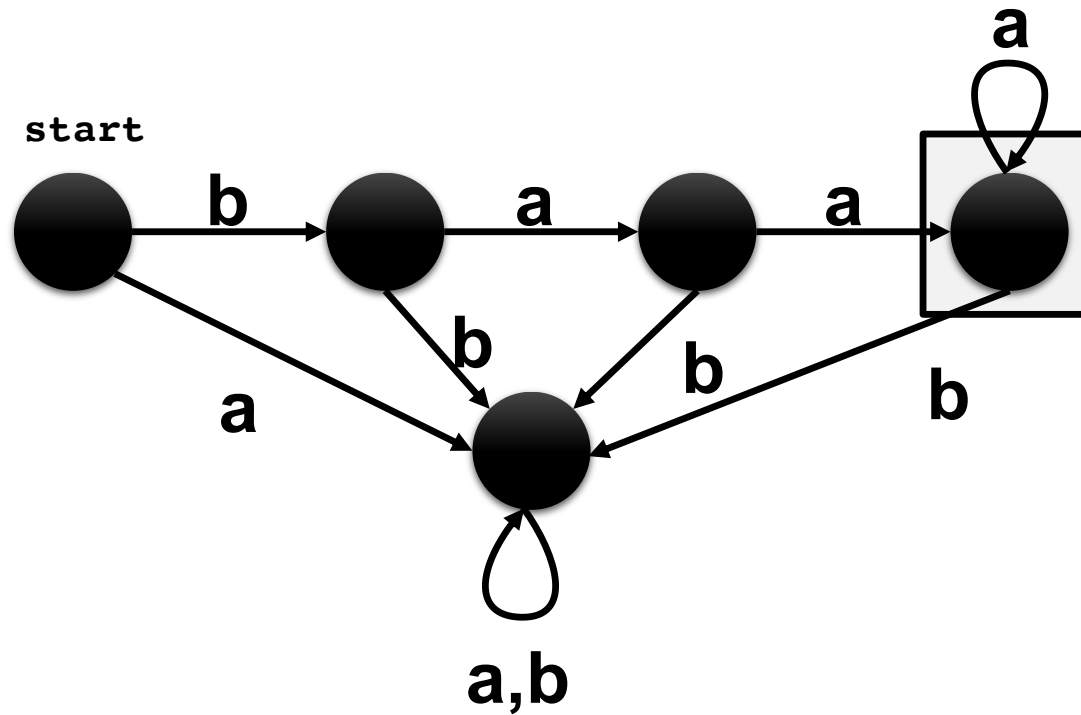
Information Retrieval

11. Language models

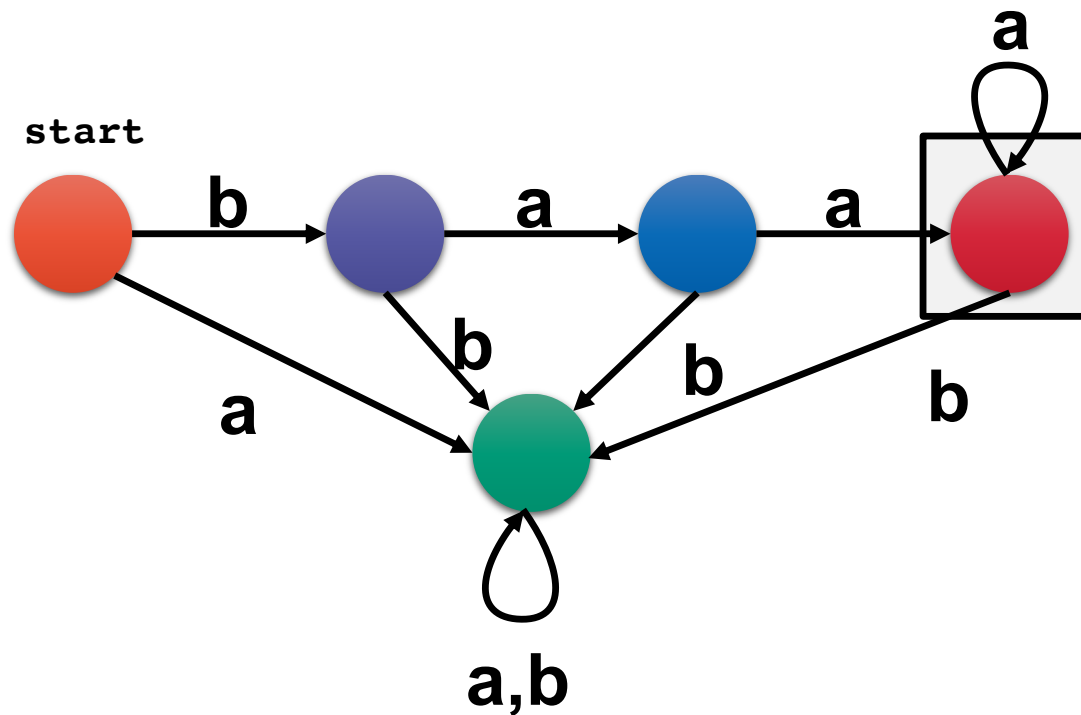
Ghislain Fourny
Spring 2022

11

Finite State Automaton (FSA)

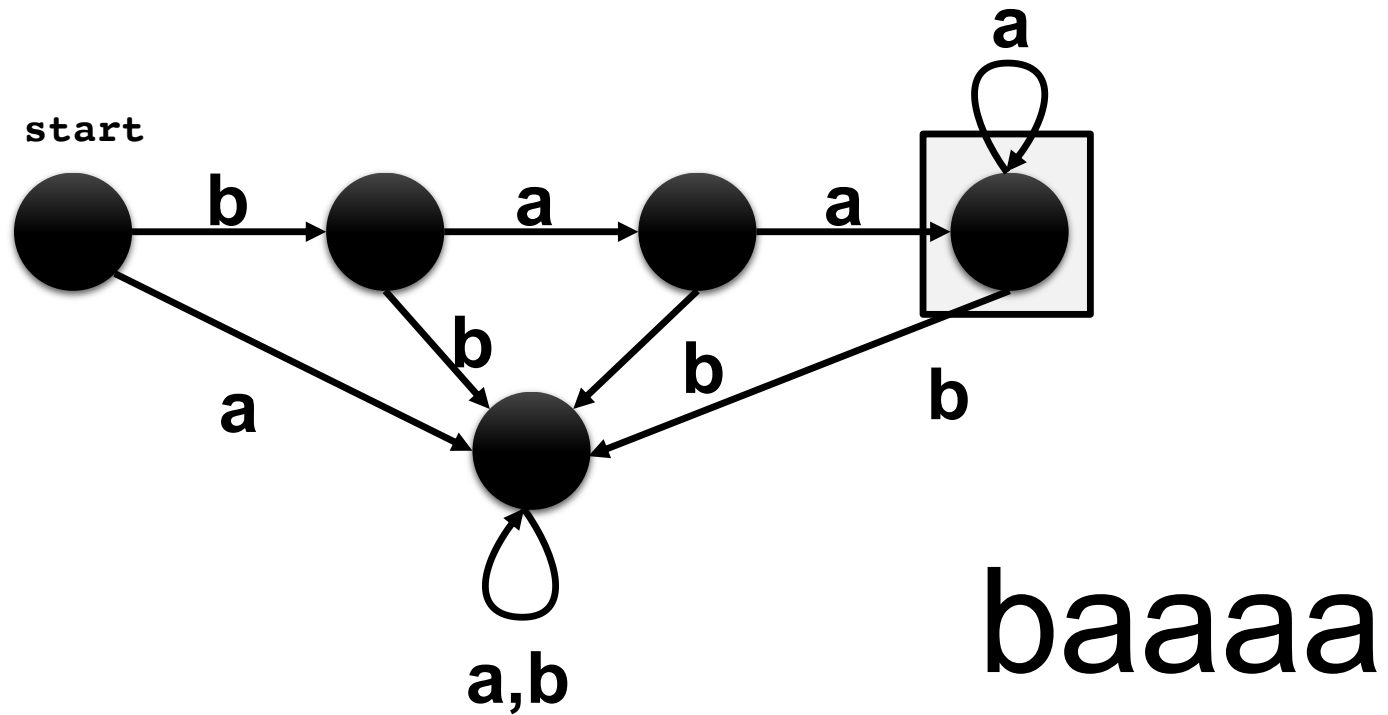


Finite State Automaton (FSA) – Transition table

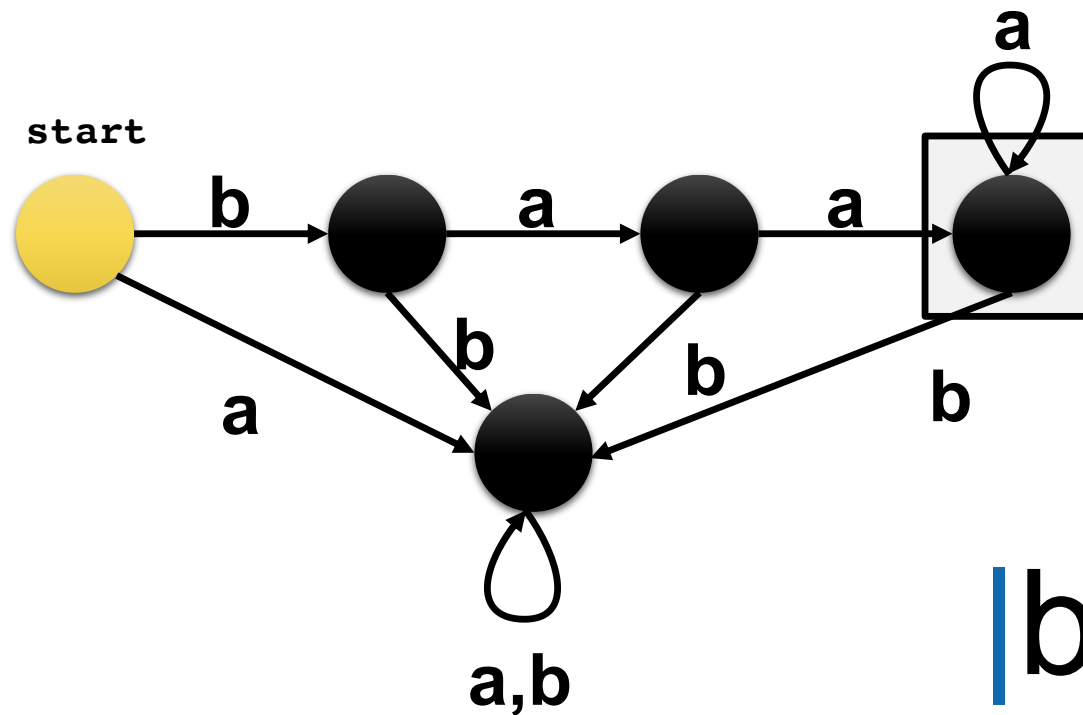


State	a	b
1	5	2
2	3	5
3	4	5
4	4	5
5	5	5

Finite State Automaton (FSA)

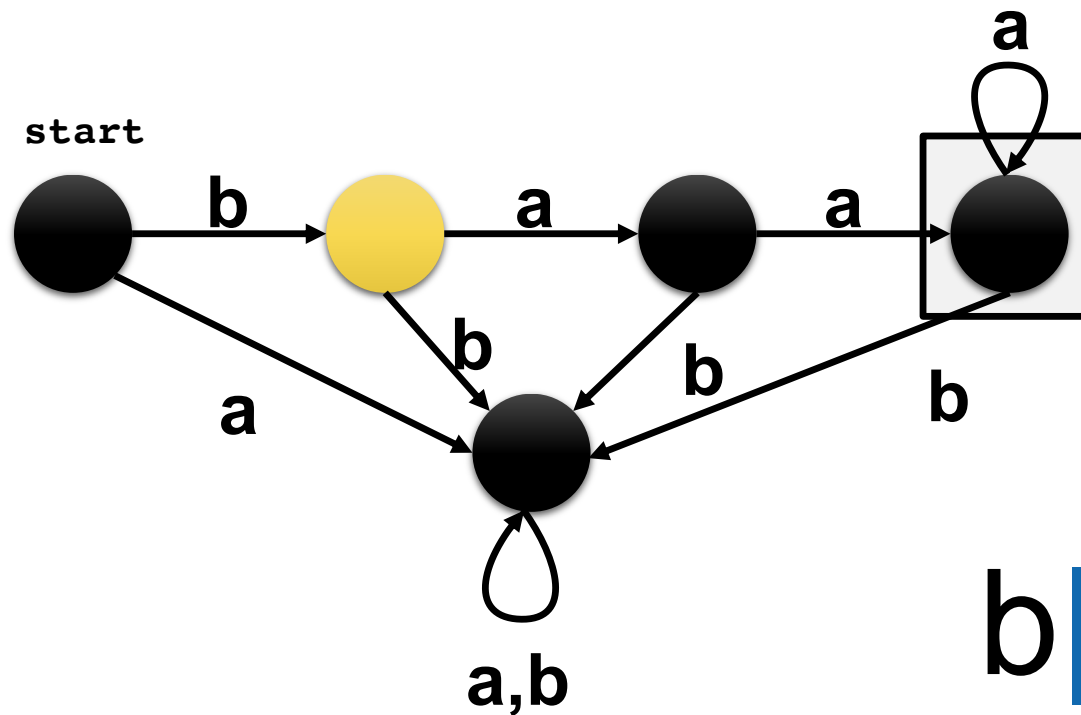


Finite State Automaton (FSA)



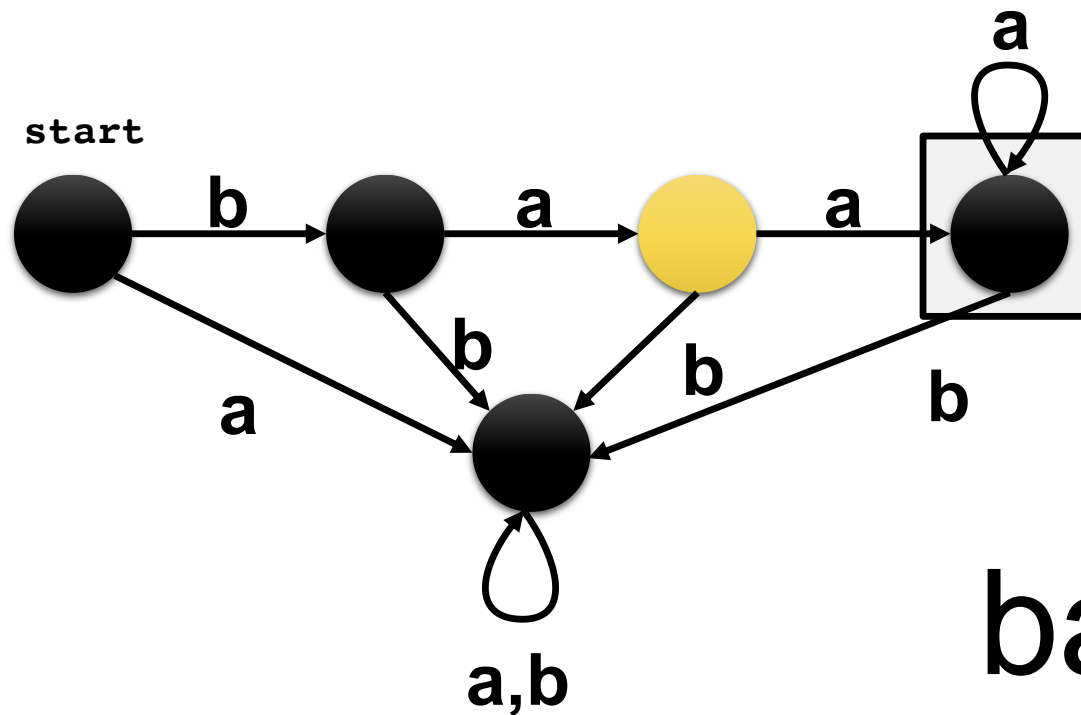
|baaaaa

Finite State Automaton (FSA)



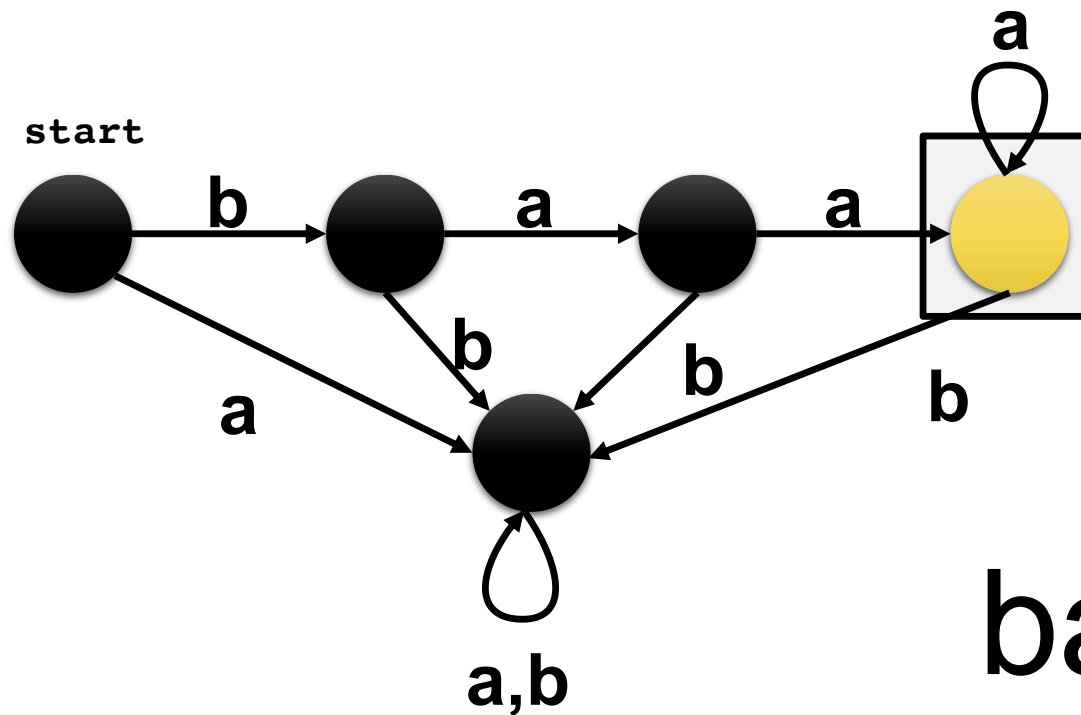
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Finite State Automaton (FSA)



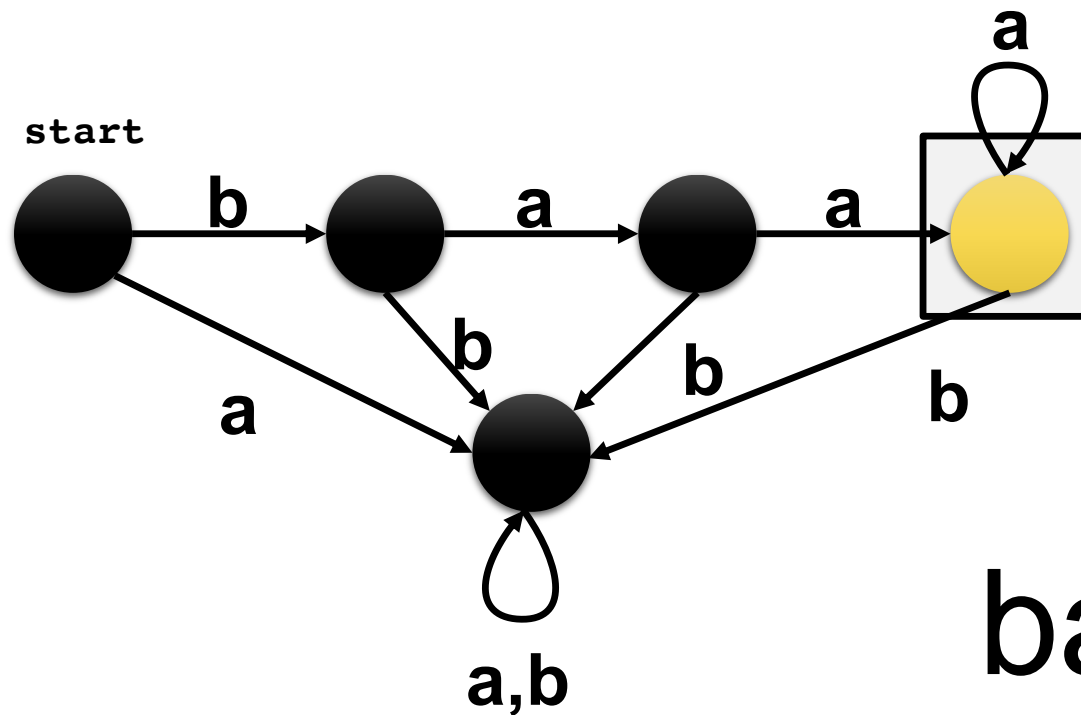
ba|aaa

Finite State Automaton (FSA)



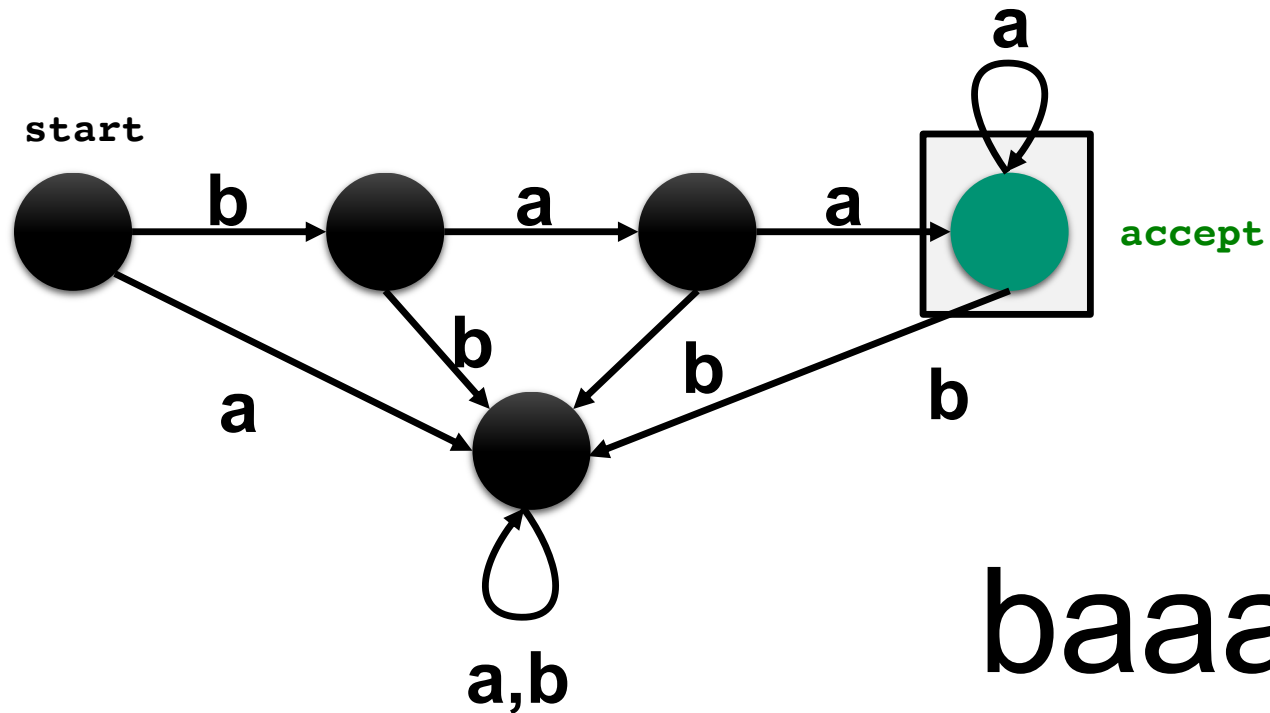
baa|aa

Finite State Automaton (FSA)

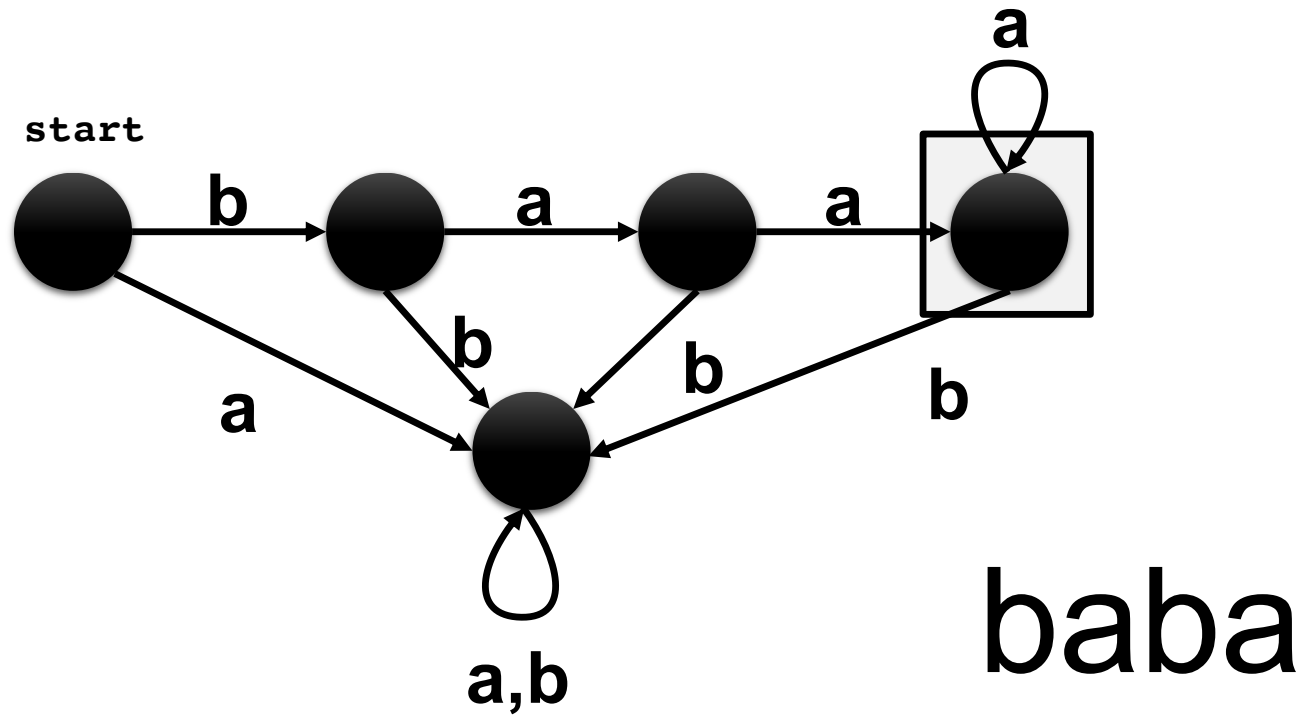


baaa|a

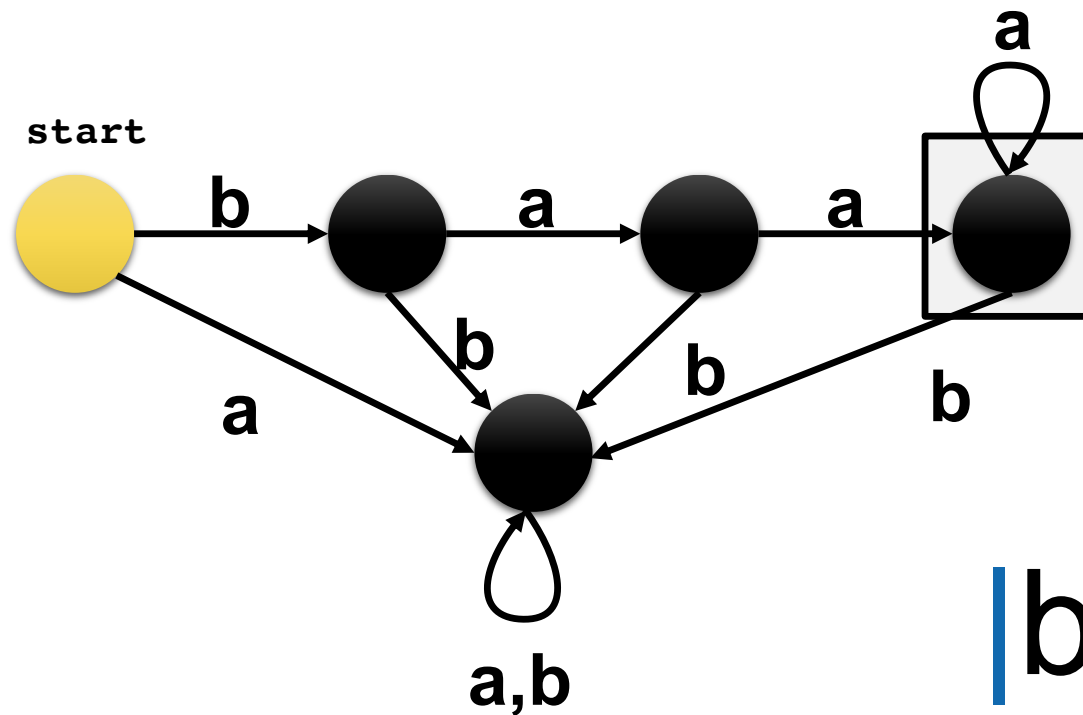
Finite State Automaton (FSA)



Finite State Automaton (FSA)

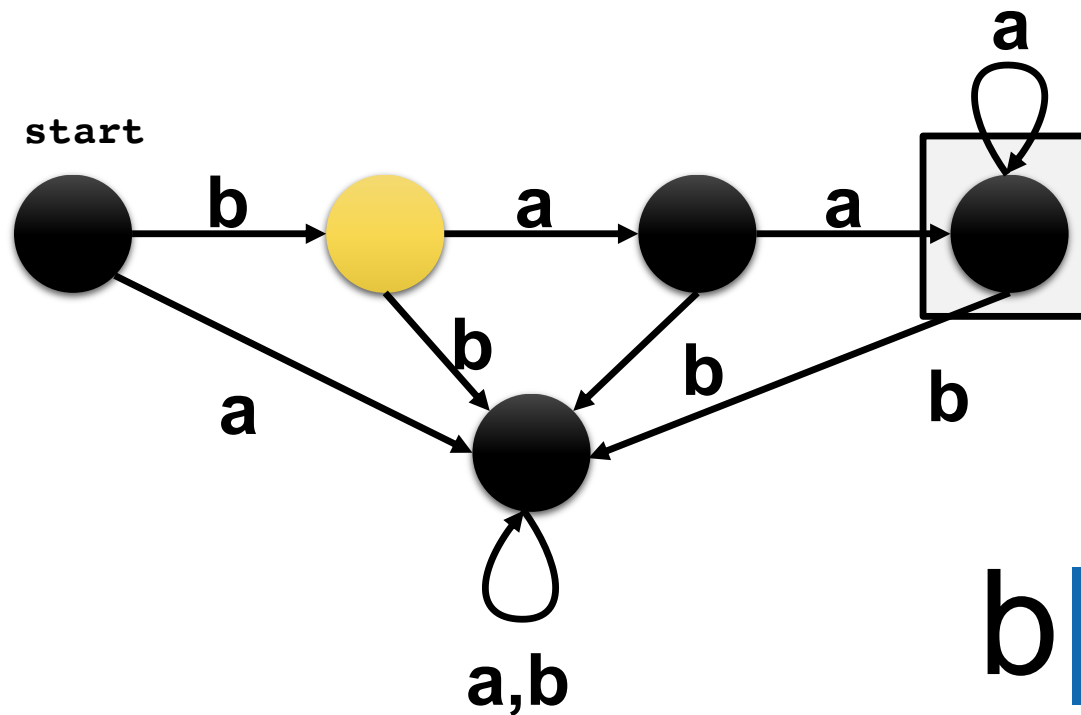


Finite State Automaton (FSA)



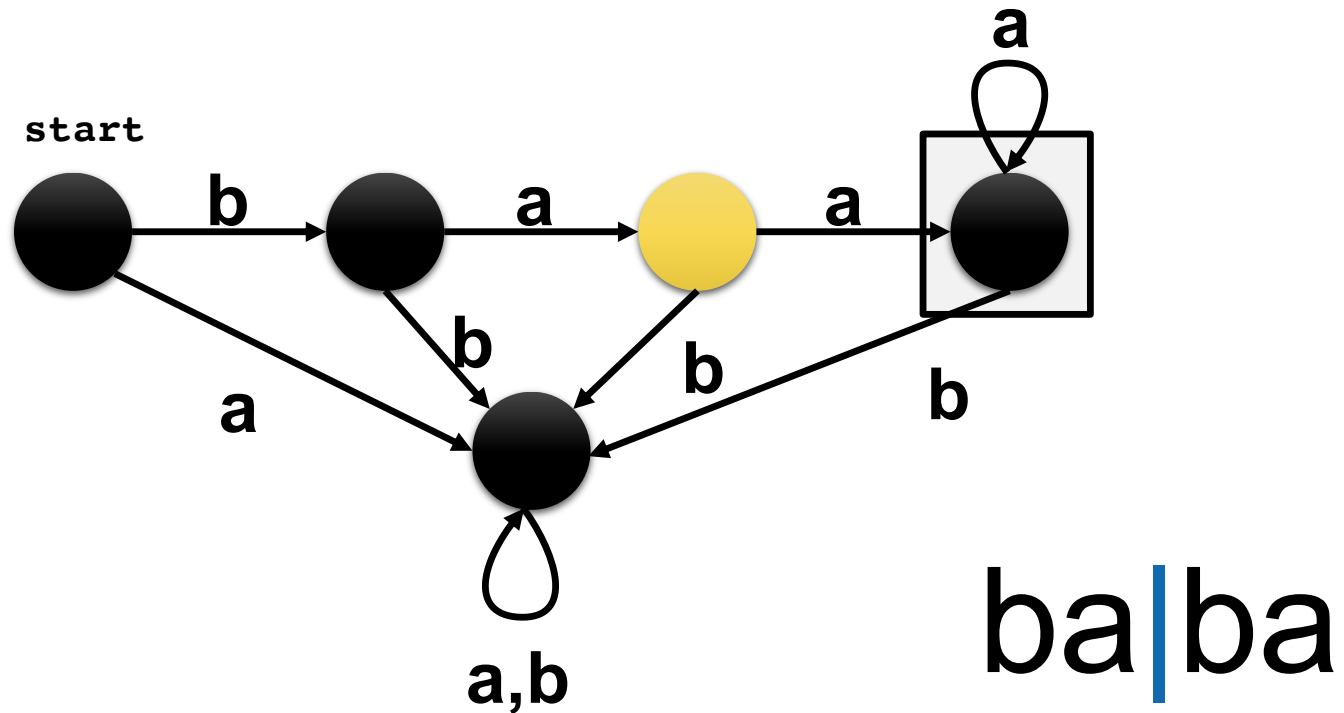
|baba

Finite State Automaton (FSA)

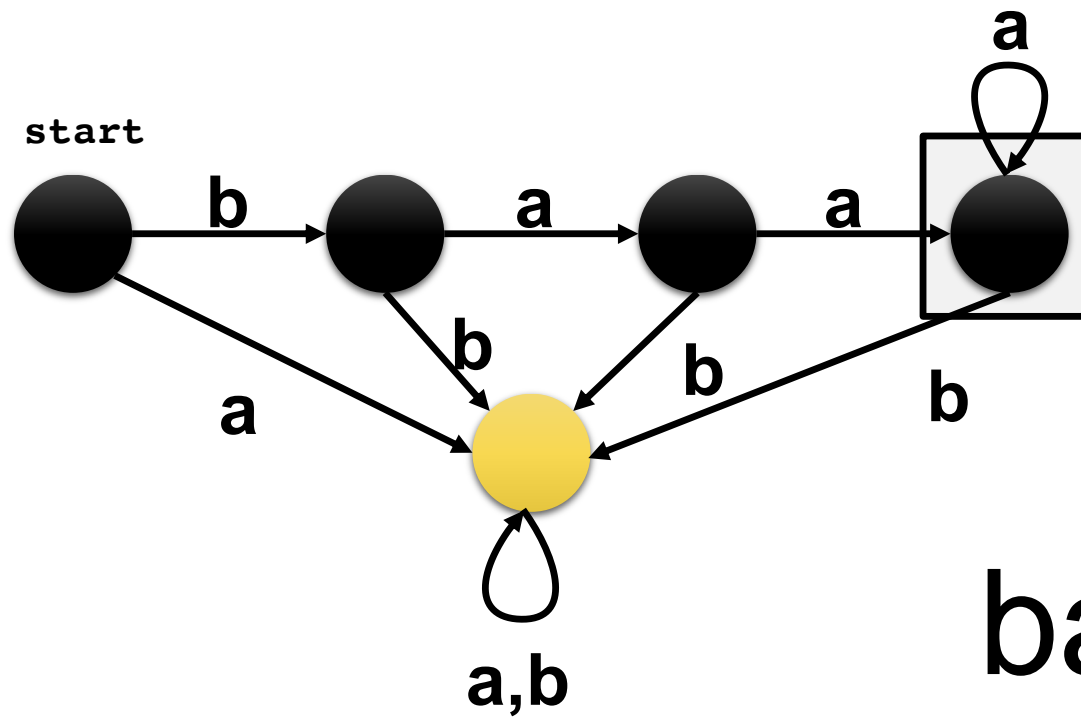


b|aba

Finite State Automaton (FSA)

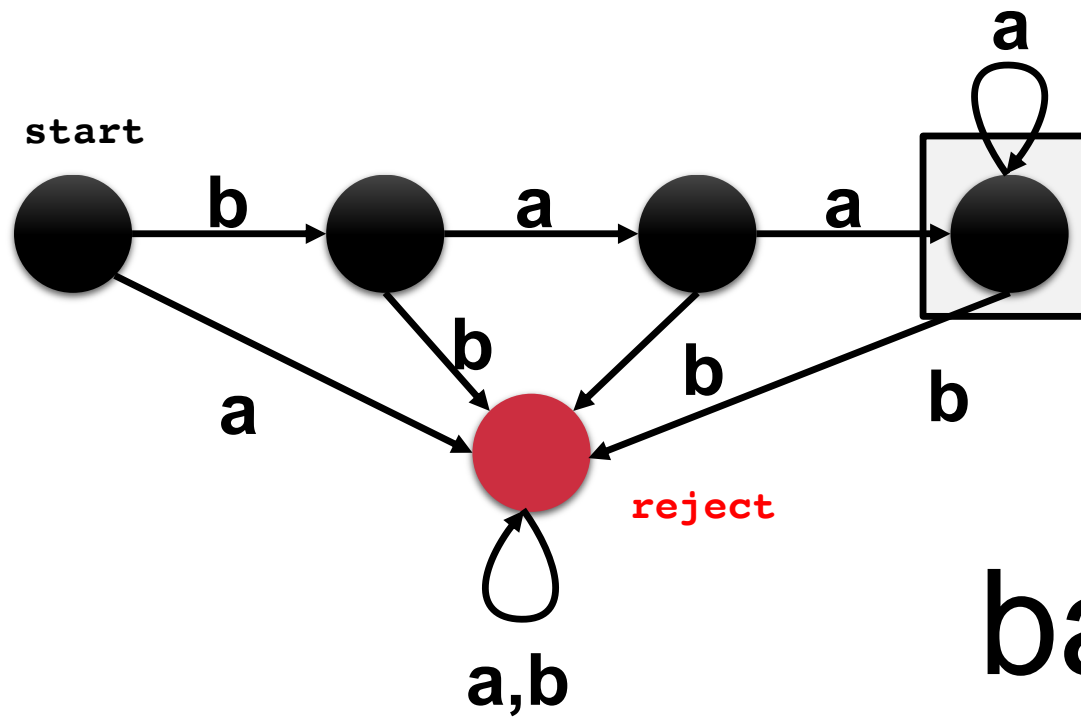


Finite State Automaton (FSA)

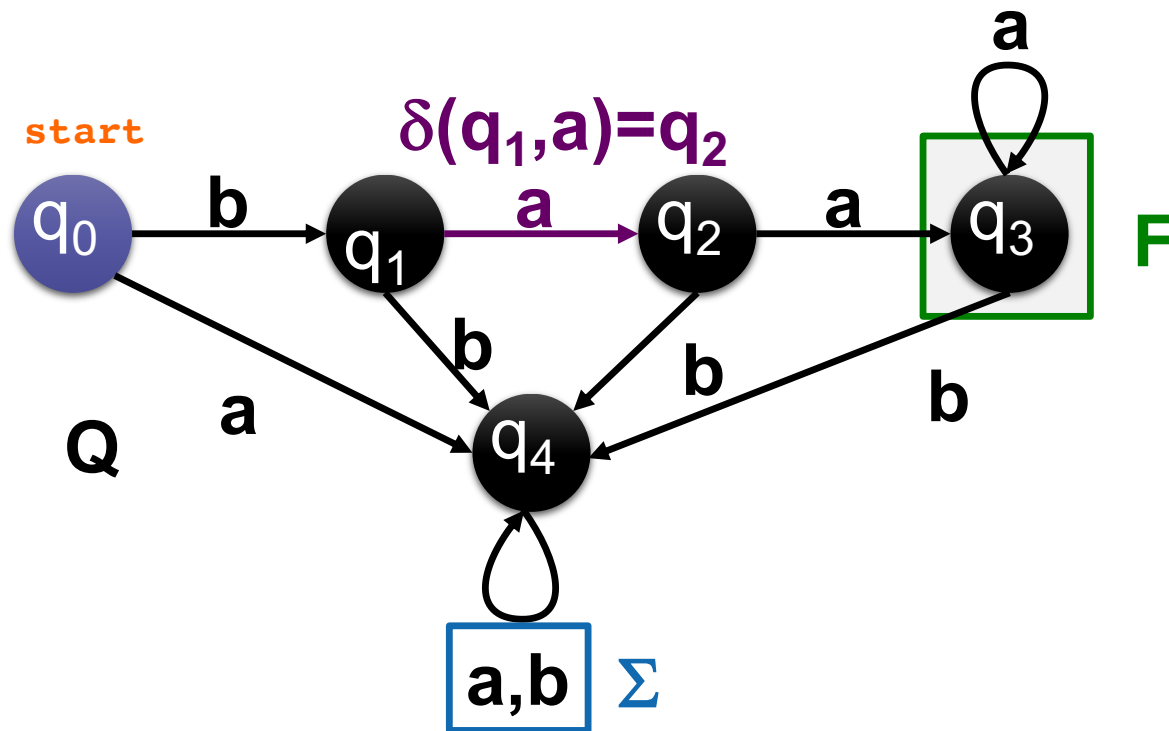


`bab|a`

Finite State Automaton (FSA)



Finite State Automaton - formally



Language model

Language space

baa
baaa
baaaa
...

$\mathcal{L}$

Language model

Language space

baa
baaa
baaaa
...

Ω

Language model

Language space

baa
baaa
baaaa
...

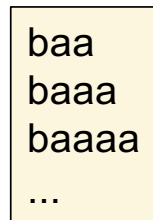
Ω

$$p : \begin{array}{l} \Omega \rightarrow [0, 1] \\ \omega \mapsto p(\omega) \end{array}$$

$$\sum_{\omega \in \Omega} p(\omega) = 1$$

Language model

Language space



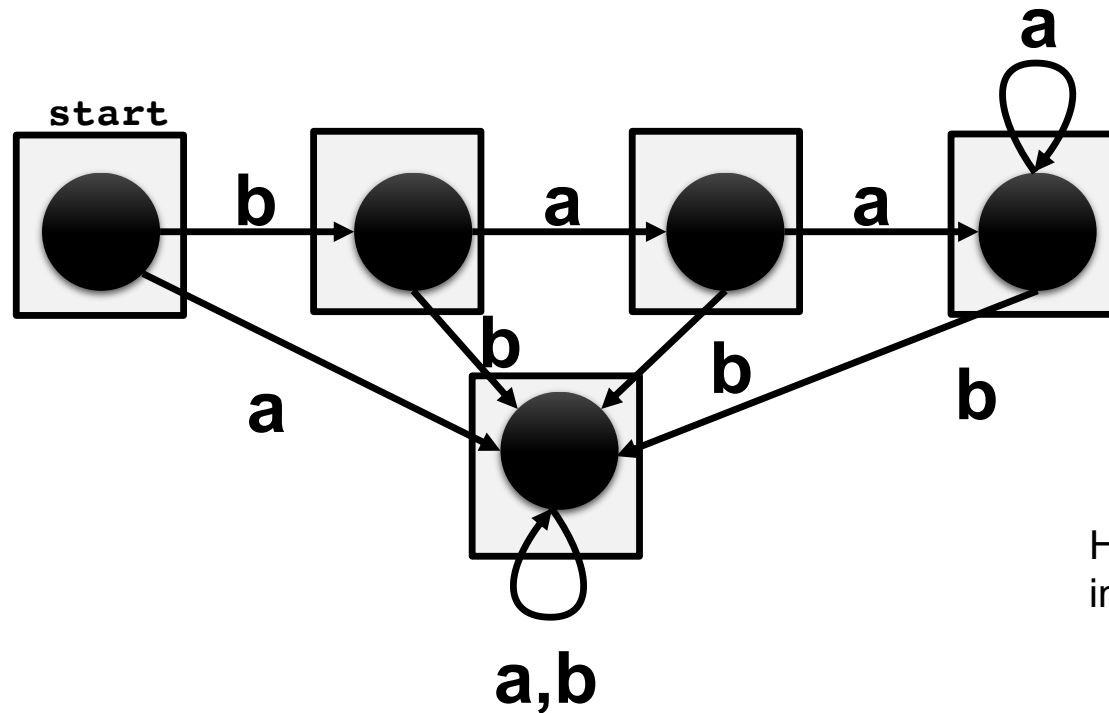
baa
baaa
baaaa
...

$\mathcal{L}$

$$p : \begin{array}{l} \mathcal{L} \rightarrow [0, 1] \\ s \mapsto p(s) \end{array}$$

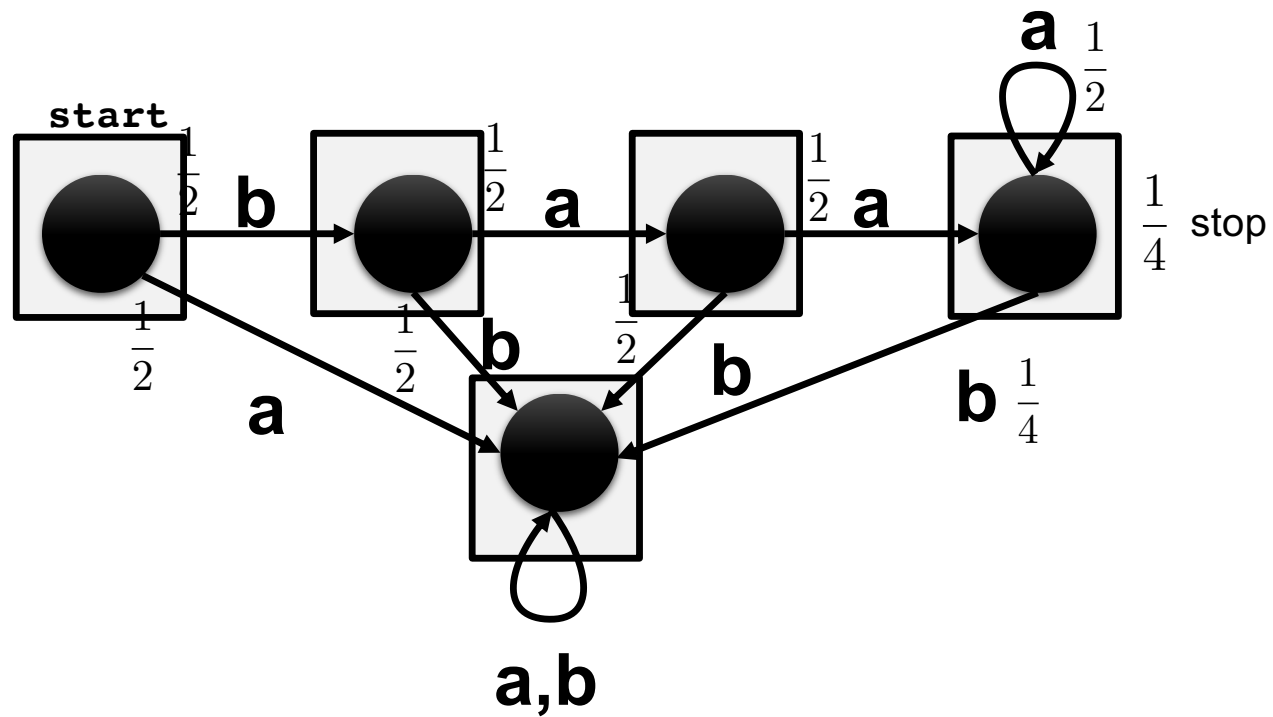
$$\sum_{s \in \mathcal{L}} p(s) = 1$$

Finite State Automaton (FSA)

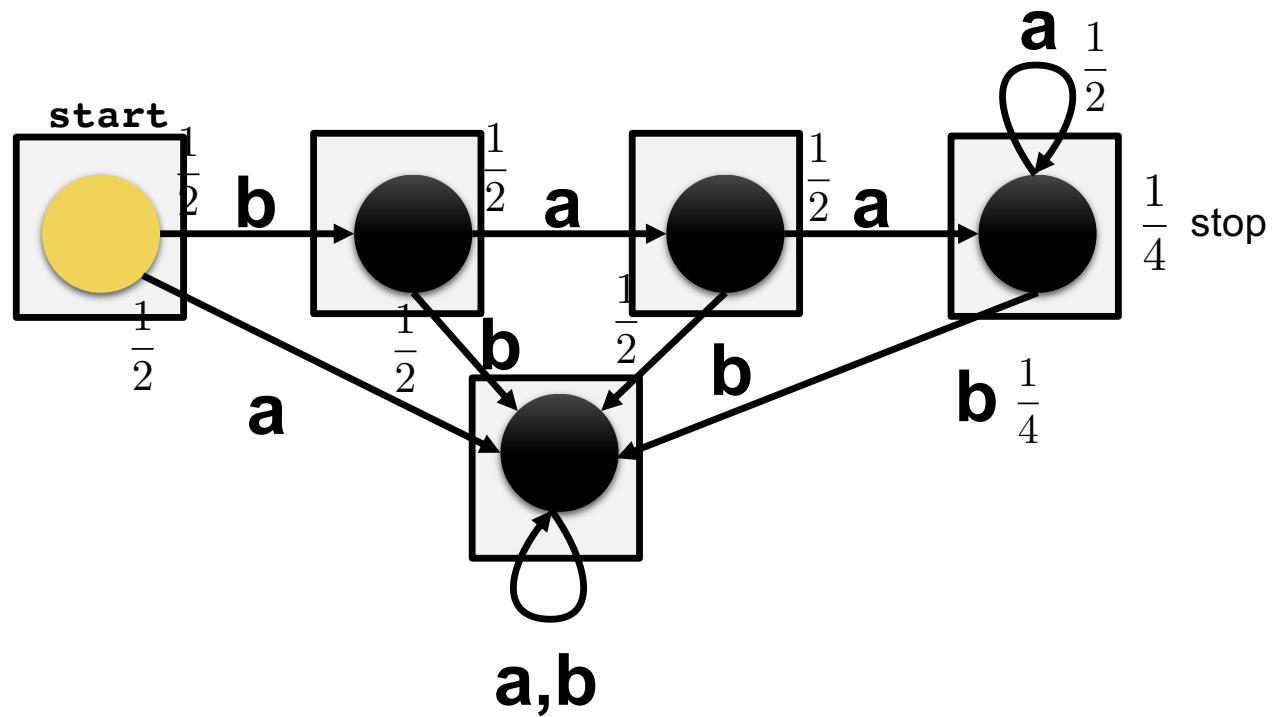


How do we turn this
into a **generator**?

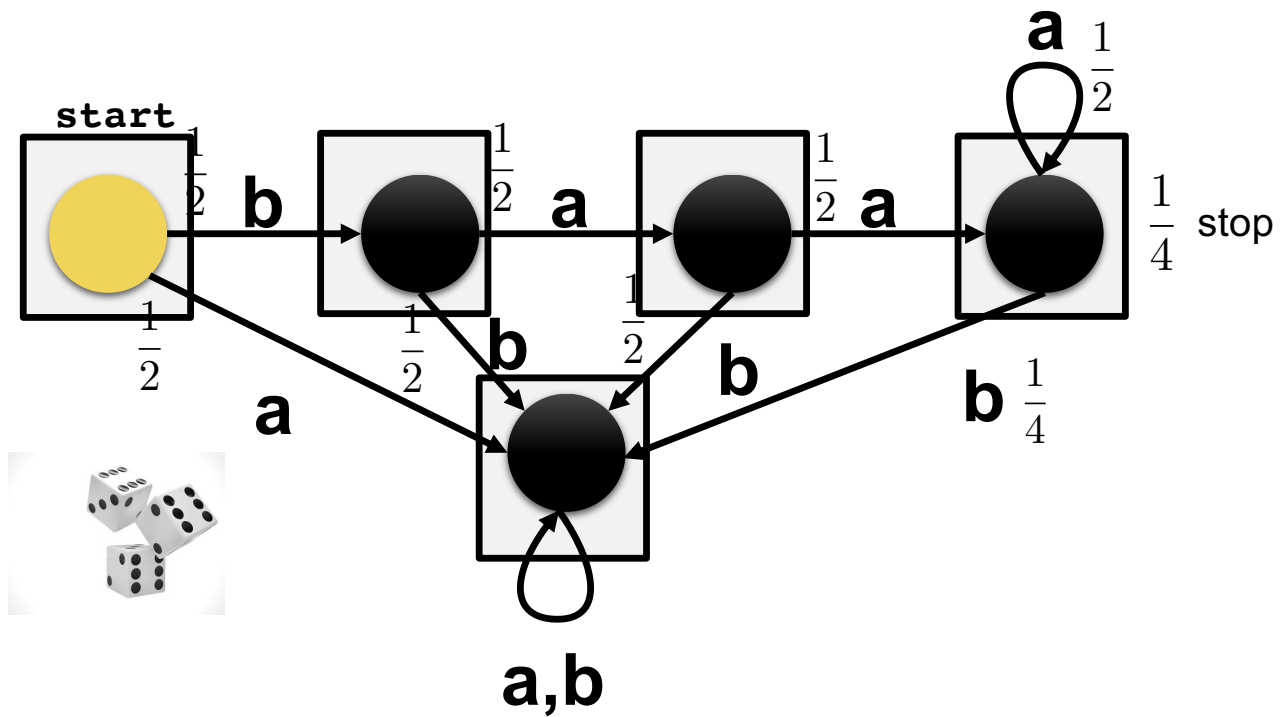
Finite State Automaton (FSA)



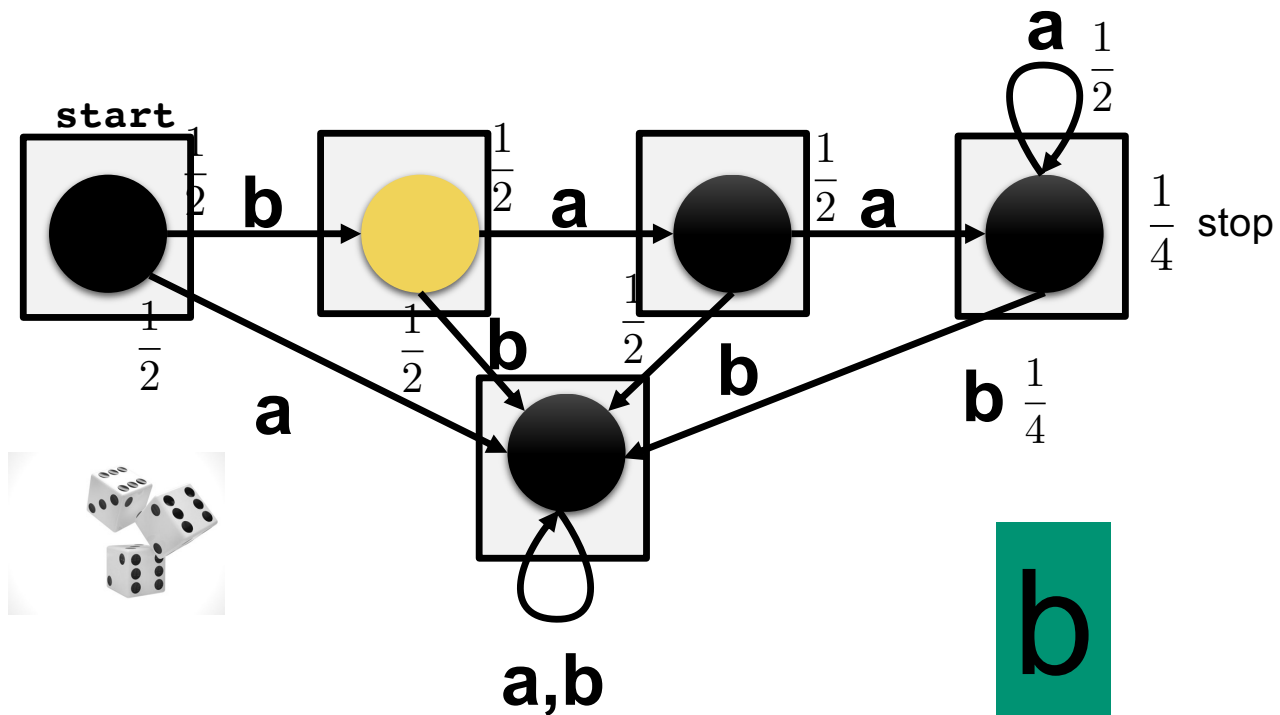
Finite State Automaton (FSA)



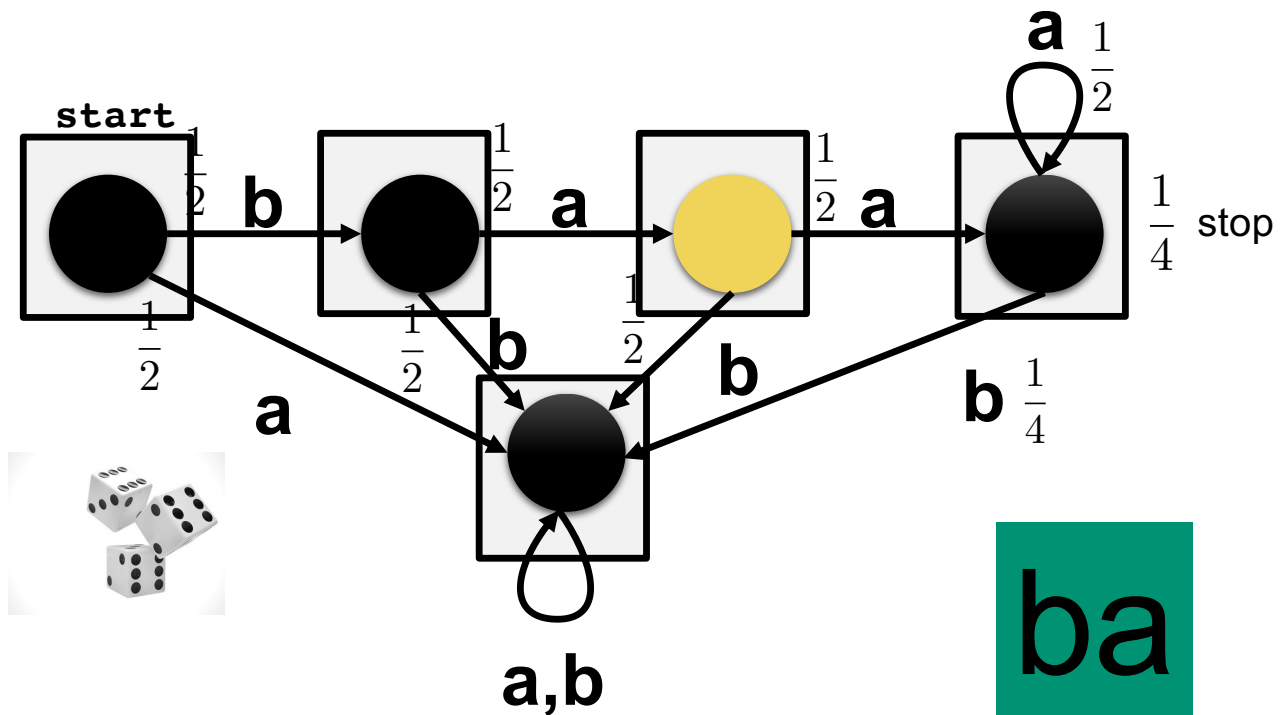
Finite State Automaton (FSA)



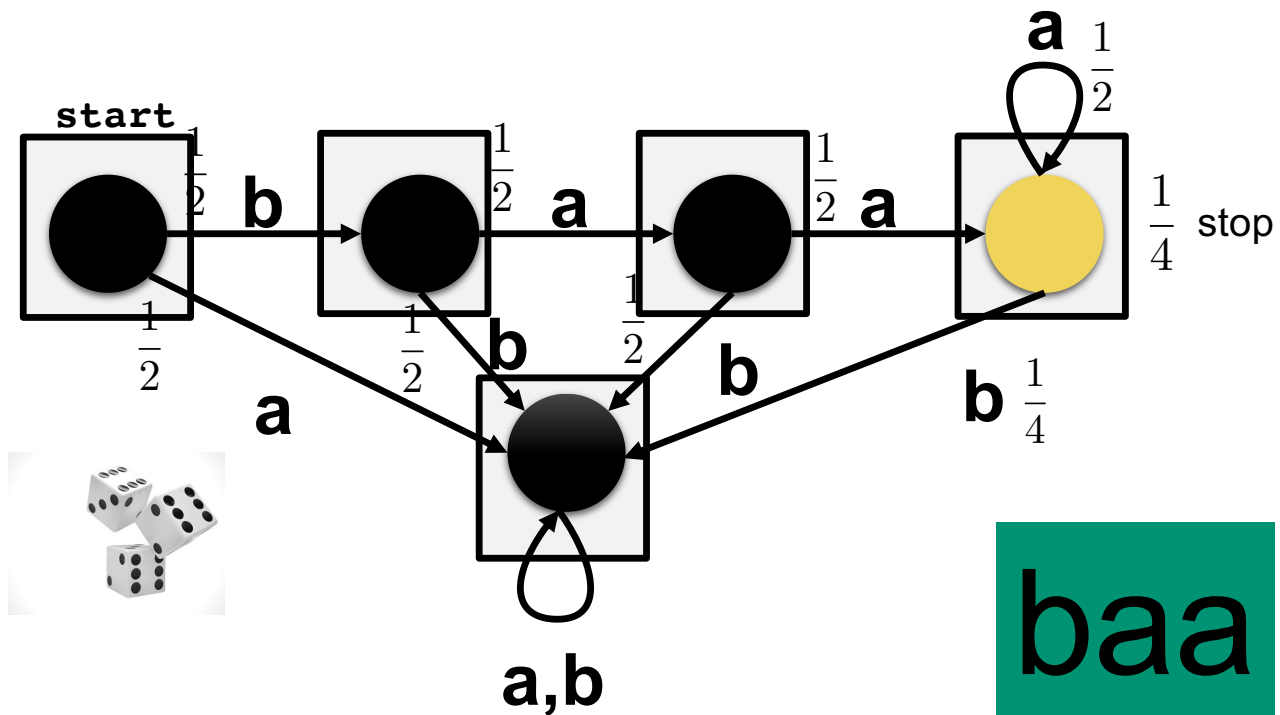
Finite State Automaton (FSA)



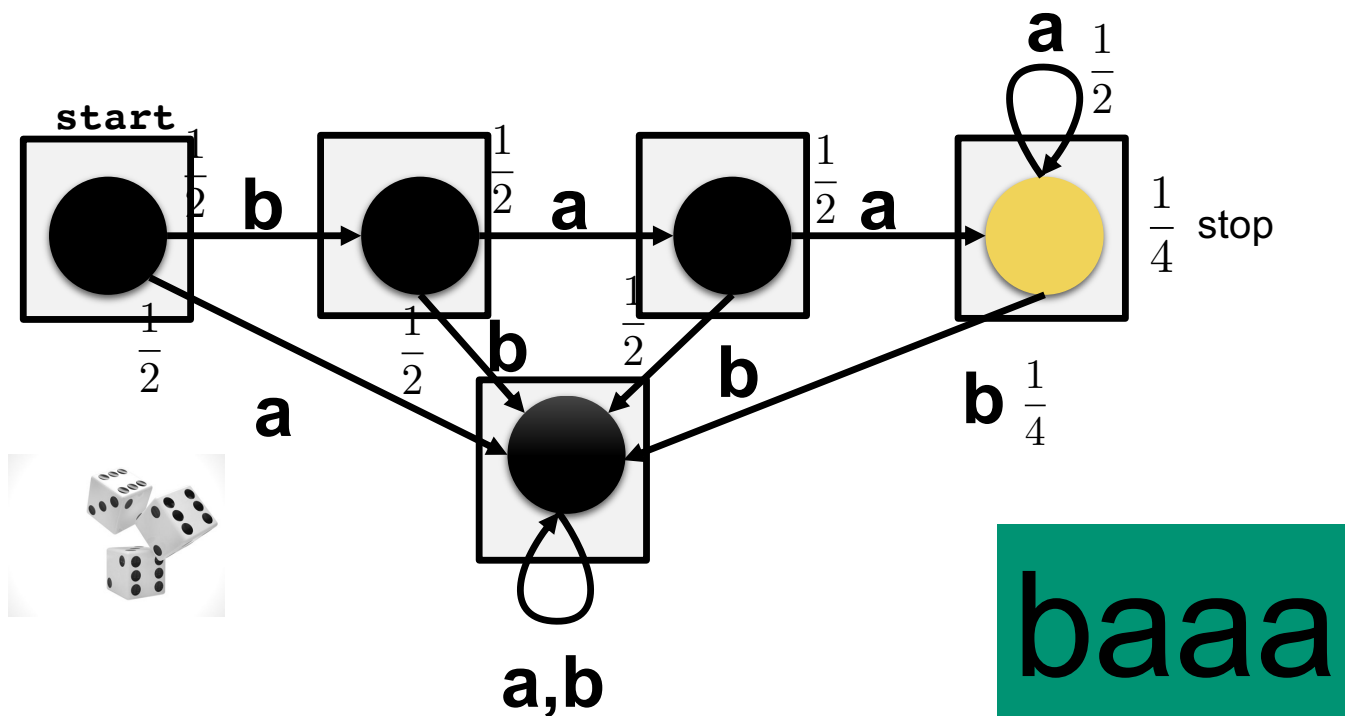
Finite State Automaton (FSA)



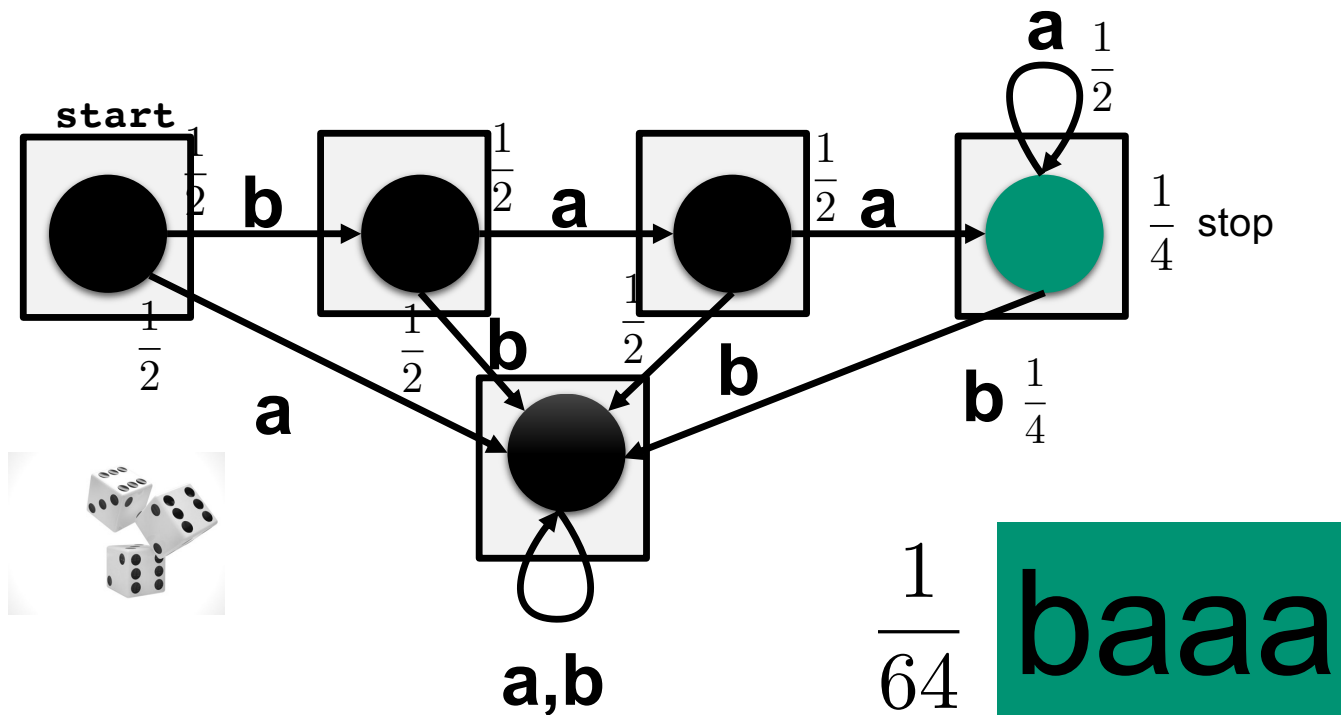
Finite State Automaton (FSA)



Finite State Automaton (FSA)



Finite State Automaton (FSA)



Generating a document at random



$$L \in \Omega \rightarrow \mathcal{T}^*$$

Random document

Generating a document at random



$L \in \Omega \rightarrow \mathcal{T}^*$ Random document

List of words model!

Generating a document at random



$$L \in \Omega \rightarrow \mathcal{T}^*$$

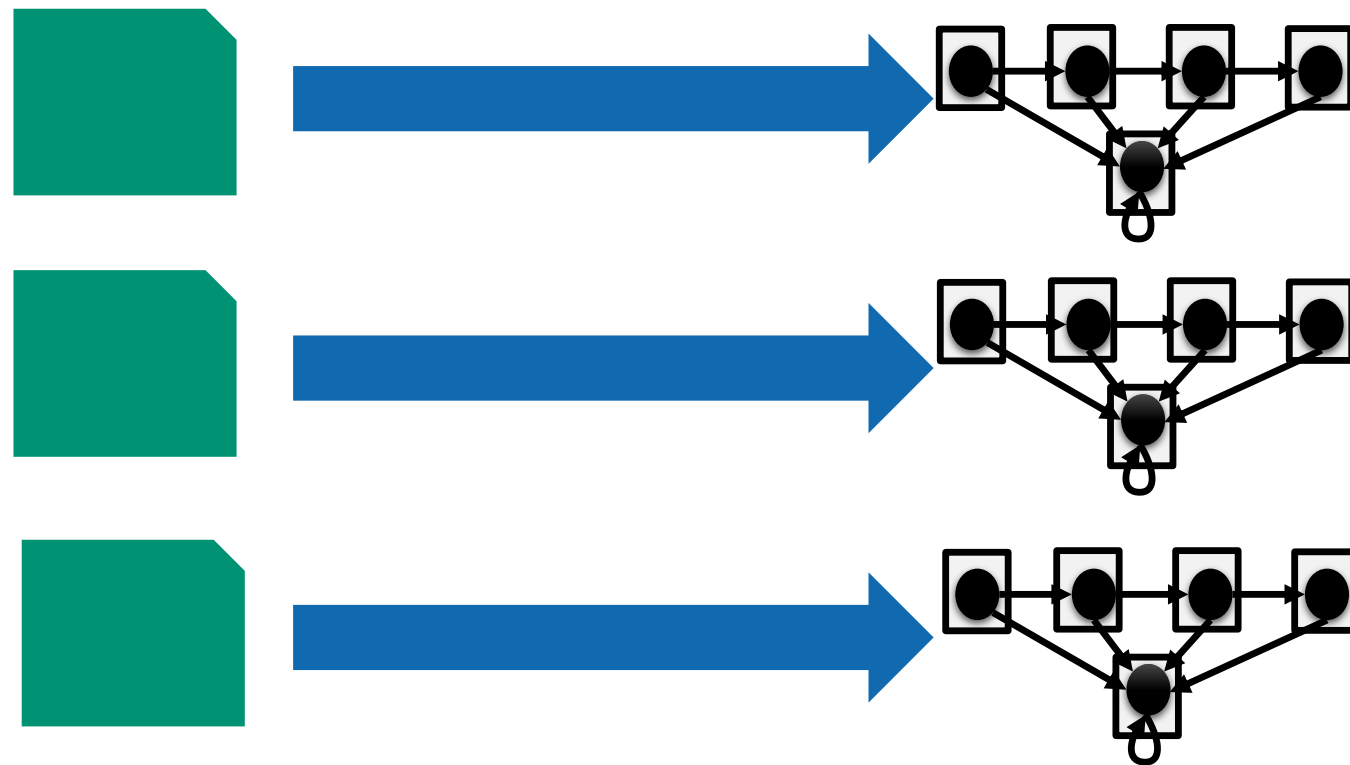
Random document



$$\|L\| \in \Omega \rightarrow \mathbb{N}$$

Random length

We can generate a model for every document



Chain rule (document of length 3)

$$\begin{aligned} P(D = (d_1, d_2, d_3)) &= P(L_D \geq 1) \times (D_1 = d_1 | L_D \geq 1) \\ &\quad \times P(L_D \geq 2 | L_D \geq 1 \wedge D_1 = d_1) \times P(D_2 = d_2 | L_D \geq 2 \wedge D_1 = d_1) \\ &\quad \times P(L_D \geq 3 | L_D \geq 2 \wedge D_1 = d_1 \wedge D_2 = d_2) \times P(D_3 = d_3 | L_D \geq 3 \wedge D_1 = d_1 \wedge D_2 = d_2) \\ &\quad \times P(L_D = 3 | L_D \geq 3 \wedge D_1 = d_1 \wedge D_2 = d_2 \wedge D_3 = d_3) \end{aligned}$$

Chain rule (document of length 3)

$$1 - p_{stop}$$

$$\begin{aligned} P(D = (d_1, d_2, d_3)) = & \boxed{P(L_D \geq 1)} \times (D_1 = d_1 | L_D \geq 1) \\ & \times \boxed{P(L_D \geq 2 | L_D \geq 1 \wedge D_1 = d_1)} \times P(D_2 = d_2 | L_D \geq 2 \wedge D_1 = d_1) \\ & \times \boxed{P(L_D \geq 3 | L_D \geq 2 \wedge D_1 = d_1 \wedge D_2 = d_2)} \times P(D_3 = d_3 | L_D \geq 3 \wedge D_1 = d_1 \wedge D_2 = d_2) \\ & \times \boxed{P(L_D = 3 | L_D \geq 3 \wedge D_1 = d_1 \wedge D_2 = d_2 \wedge D_3 = d_3)} \end{aligned}$$

$$p_{stop}$$

(n+1)-gram: Markov rule

$$P(D_k = d_k \mid L_D \geq k \wedge D_1 = d_1 \wedge D_2 = d_2 \wedge \dots \wedge D_{k-1} = d_{k-1})$$

=

$$P(D_k = d_k \mid L_D \geq k \wedge \underbrace{D_{k-n} = d_{k-n} \wedge \dots \wedge D_{k-1} = d_{k-1}}_{\text{Only look at the previous n terms!}})$$

Example: 3-grams, document of length 4

$$\begin{aligned} P(D = (d_1, d_2, d_3, d_4)) = & \\ & (1 - p_{stop})P(D_1 = d_1) \\ & (1 - p_{stop})P(D_2 = d_2|D_1 = d_1) \\ & (1 - p_{stop})P(D_3 = d_3|D_1 = d_1 \wedge D_2 = d_2) \\ & (1 - p_{stop})P(D_4 = d_4|D_2 = d_2 \wedge D_3 = d_3)p_{stop} \end{aligned}$$



Unigrams

1-gram: Markov rule

$$P(D_k = d_k \mid L_D \geq k \wedge D_1 = d_1 \wedge D_2 = d_2 \wedge \dots \wedge D_{k-1} = d_{k-1})$$

$$\begin{aligned} &= \\ P(D_k = d_k \mid L_D \geq k) \\ &= \\ p_{d_k} \end{aligned}$$

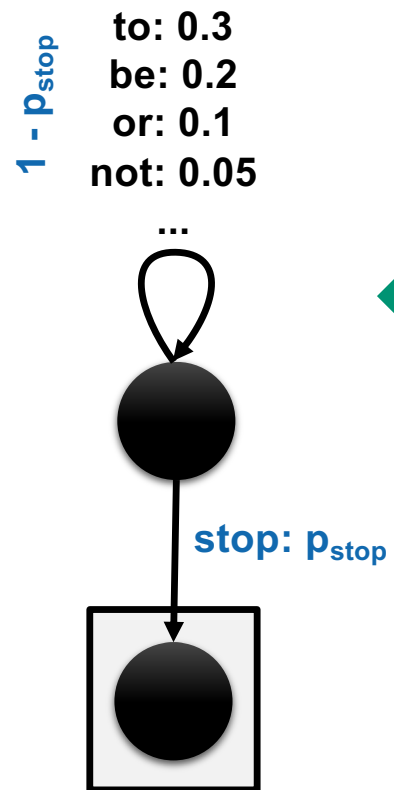
How?



Term	p
to	0.3
be	0.2
or	0.1
not	0.05
that	0.15
is	0.1

Unigram model

Corresponding automaton (unigram model)



Term	p
to	0.3
be	0.2
or	0.1
not	0.05
that	0.15
is	0.1

Unigram model

Generating a document at random

$$P(D = (d_1, d_2, d_3)) = ?$$

Unigram (no memory)

$$P(D = (d_1, d_2, d_3)) = (1 - p_{stop})p_{d_1}(1 - p_{stop})p_{d_2}(1 - p_{stop})p_{d_3}p_{stop}$$

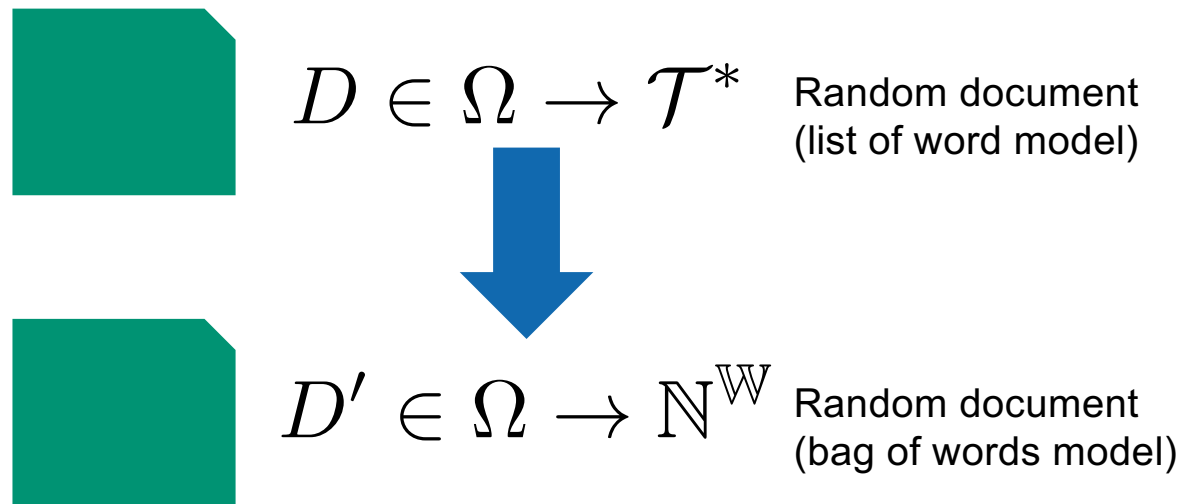
How do we get rid of the order?



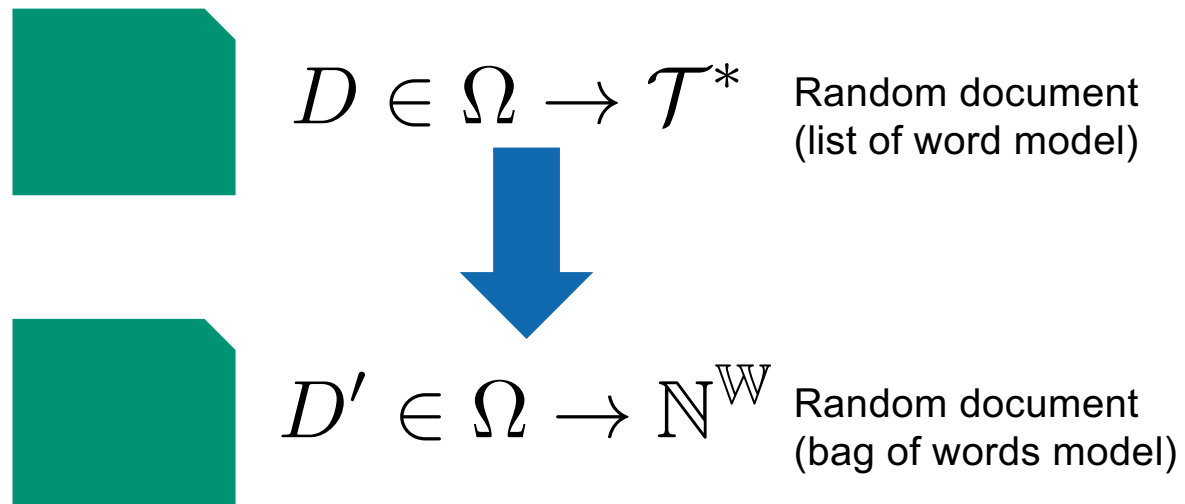
$$D \in \Omega \rightarrow \mathcal{T}^*$$

Random document
(list of word model)

How do we get rid of the order?

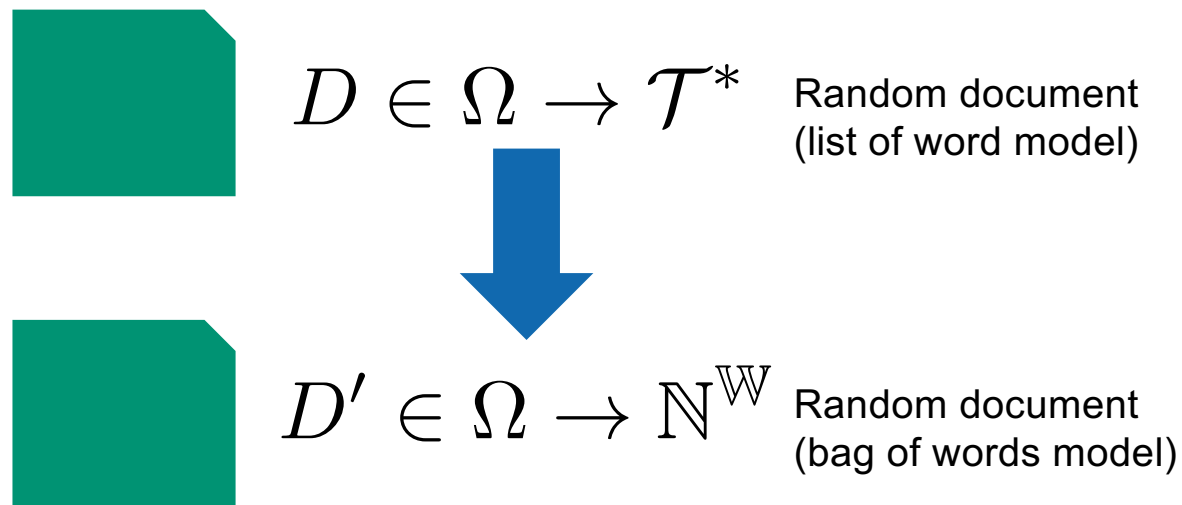


How do we get rid of the order?



$$P(D' = d') = \sum_{d \text{ matching bag } d'} P(D = d)$$

How do we get rid of the order?



$$P(D' = d') = \sum_{d \text{ matching bag } d'} P(D = d)$$

(This is actually a multinomial distribution: see combinatorics)

Multinomial distribution

$$P(D = d) = \frac{L_d!}{\prod_t {}^t f_{t,d}!} \prod_t p_t^{{}^t f_{t,d}} (1 - p_{stop})^{L_d} p_{stop}$$



K-grams

Bigram



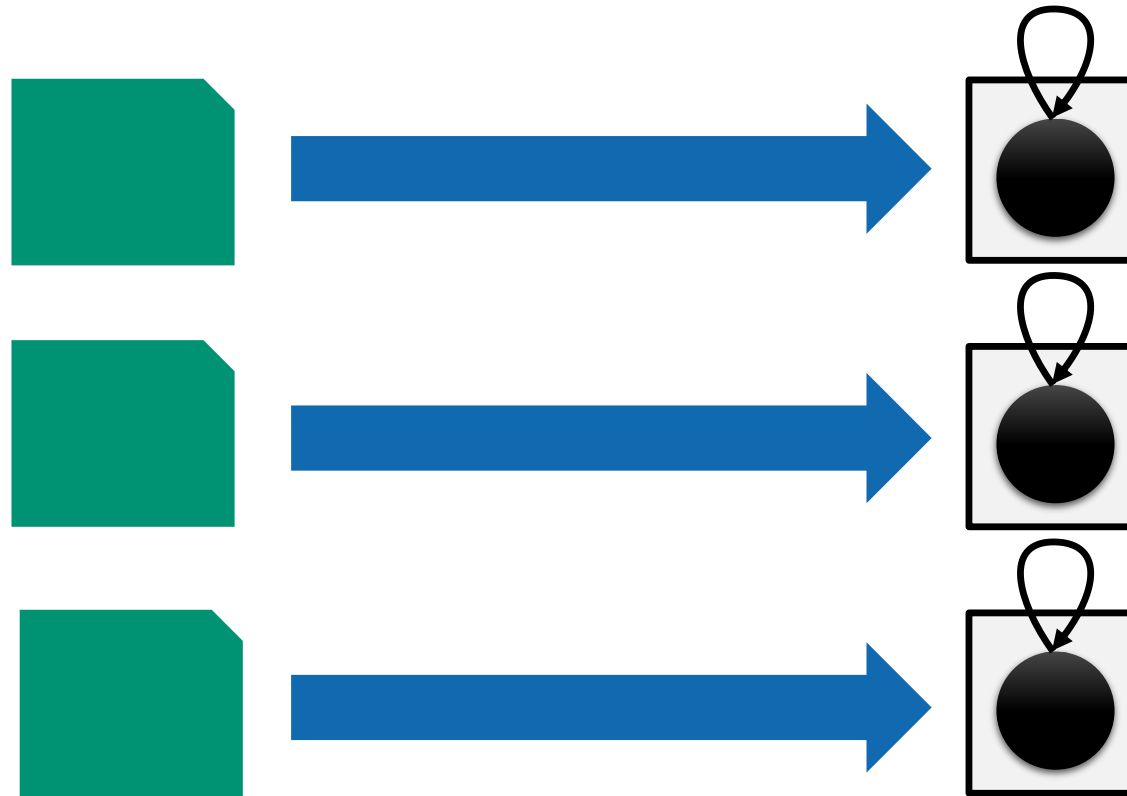
Previous term	Current term	p
to	to	0.3
to	be	0.2
to	or	0.5
be	to	0.05
be	be	0.25
be	or	0.75

Bigram model

Bigram

$$P(D = (d_1, d_2, d_3)) = (1 - p_{stop})p_{d_1}(1 - p_{stop})p_{d_1, d_2}(1 - p_{stop})p_{d_2, d_3}p_{stop}$$

We generate a model for every document



Generating text with K-gram models

1 gram	–To him swallowed confess hear both. Which. Of save on trail for are ay device and rote life have –Hill he late speaks; or! a more to leg less first you enter
2 gram	–Why dost stand forth thy canopy, forsooth; he is this palpable hit the King Henry. Live king. Follow. –What means, sir. I confess she? then all sorts, he is trim, captain.
3 gram	–Fly, and will rid me these news of price. Therefore the sadness of parting, as they say, 'tis done. –This shall forbid it should be branded, if renown made it empty.
4 gram	–King Henry. What! I will go seek the traitor Gloucester. Exeunt some of the watch. A great banquet serv'd in; –It cannot be but so.

Figure 3.3 Eight sentences randomly generated from four n-grams computed from Shakespeare's works. All characters were mapped to lower-case and punctuation marks were treated as words. Output is hand-corrected for capitalization to improve readability.

Source: Speech and Language Processing. Daniel Jurafsky & James H. Martin

Now, let's get back to information retrieval

Enters a query q



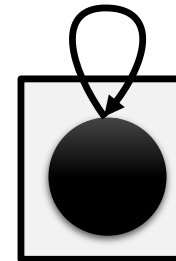
Now, let's get back to information retrieval

Enters a query q



Thought experiment: imagine that:

- we picked a random document and built its model



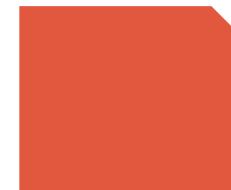
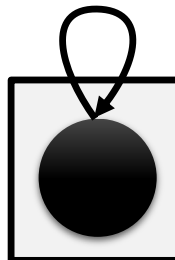
Now, let's get back to information retrieval

Enters a query q



Thought experiment: imagine that:

- we picked a random document and built its model
- we used this model to generate a new document



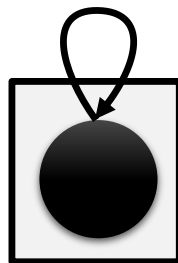
Now, let's get back to information retrieval

Enters a query q



Thought experiment: imagine that:

- we picked a random document and built its model
- we used this model to generate a new document
- that document turns out to be q



Now, let's get back to information retrieval

Enters a query q

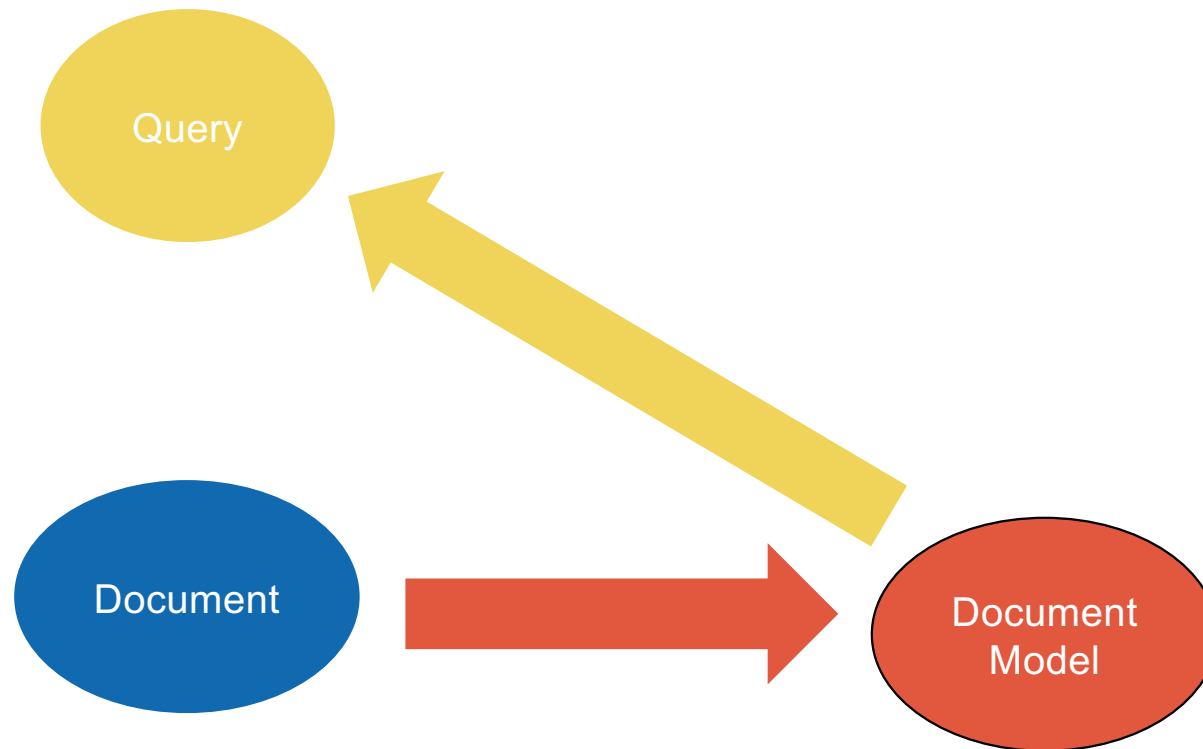


Thought experiment: imagine that:

- we picked a random document and built its model
- we used this model to generate a new document
- that document turns out to be q

What document is the **most likely** to have been picked and to have generated q ?

Query likelihood model



Bayesian model

$$P(D = d|Q = q) = \frac{P(Q = q|D = d).P(D = d)}{P(Q = q)}$$

Bayesian model

We need to sort by this

$$P(D = d|Q = q) = \frac{P(Q = q|D = d).P(D = d)}{P(Q = q)}$$

Bayesian model

$$P(D = d|Q = q) = \frac{P(Q = q|D = d).P(D = d)}{P(Q = q)}$$

We can ignore this (constant)

Bayesian model

We can also ignore this (uniform)

$$P(D = d|Q = q) = \frac{P(Q = q|D = d) \cdot \boxed{P(D = d)}}{P(Q = q)}$$

Bayesian model

This is just the probability of q
under the model built from d !

$$P(D = d|Q = q) = \frac{P(Q = q|D = d).P(D = d)}{P(Q = q)}$$

Multinomial distribution

$$P(Q = q | D = d) = \frac{L_q!}{\prod_t t f_{t,q}!} \prod_t p_t^{t f_{t,q}} (1 - p_{stop})^{L_q} p_{stop}$$

Multinomial distribution

$$P(Q = q | D = d) = \frac{L_q!}{\prod_t {}^t f_{t,q}!} \prod_t \left(\frac{{}^t f_{t,d}}{L_d} \right)^{{}^t f_{t,q}} (1 - p_{stop})^{L_q} p_{stop}$$

Smoothing

Term
Frequency

Inverted
Document
Frequency

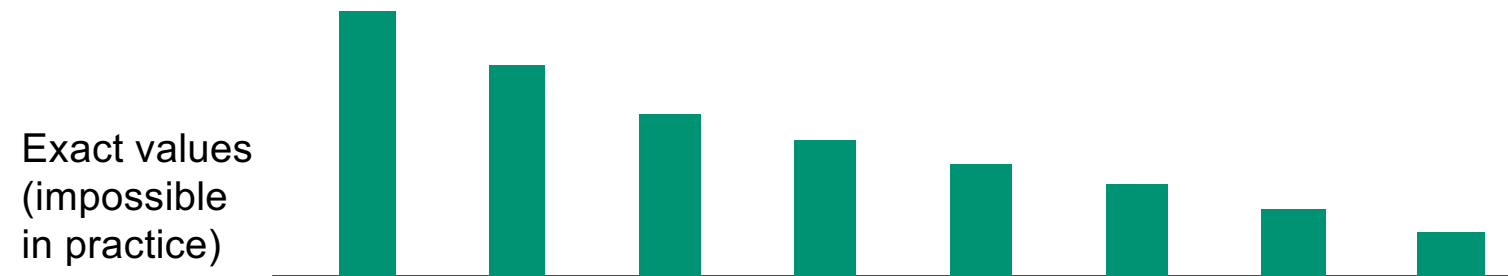
tf-idf

Language models

Probabilistic models

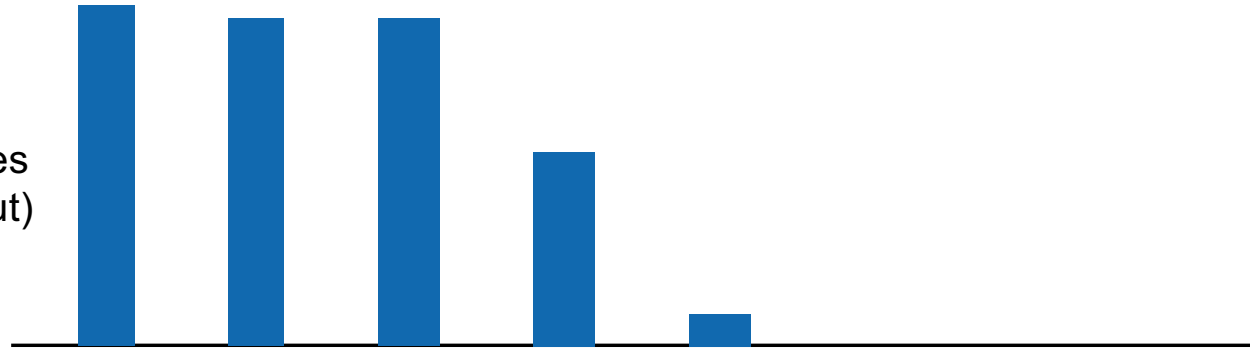
Vector space model

Reminder: smoothing

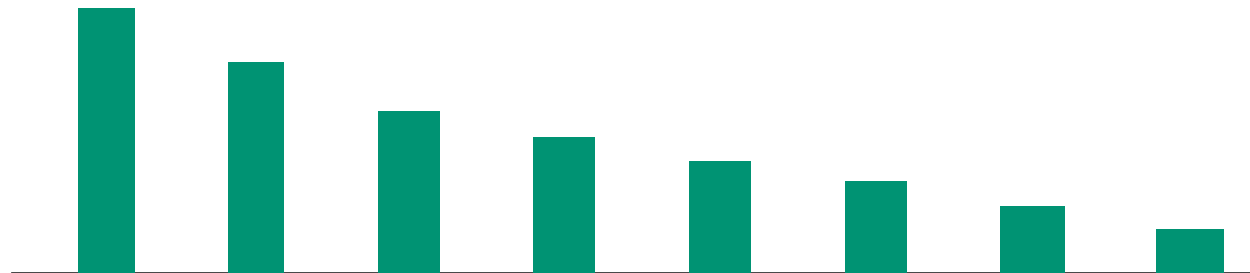


Reminder: smoothing

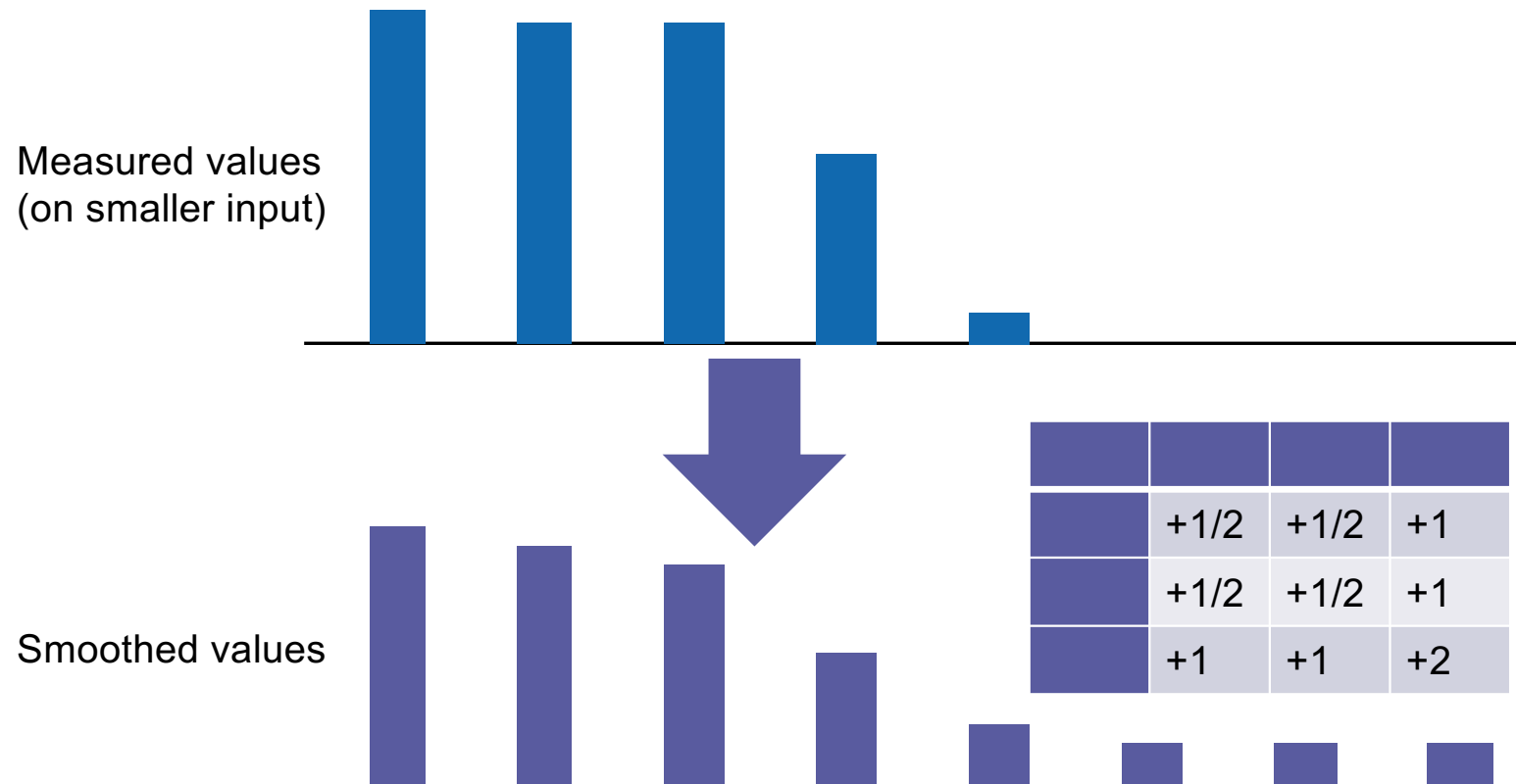
Measured values
(on smaller input)



Exact values
(impossible
in practice)



Reminder: smoothing



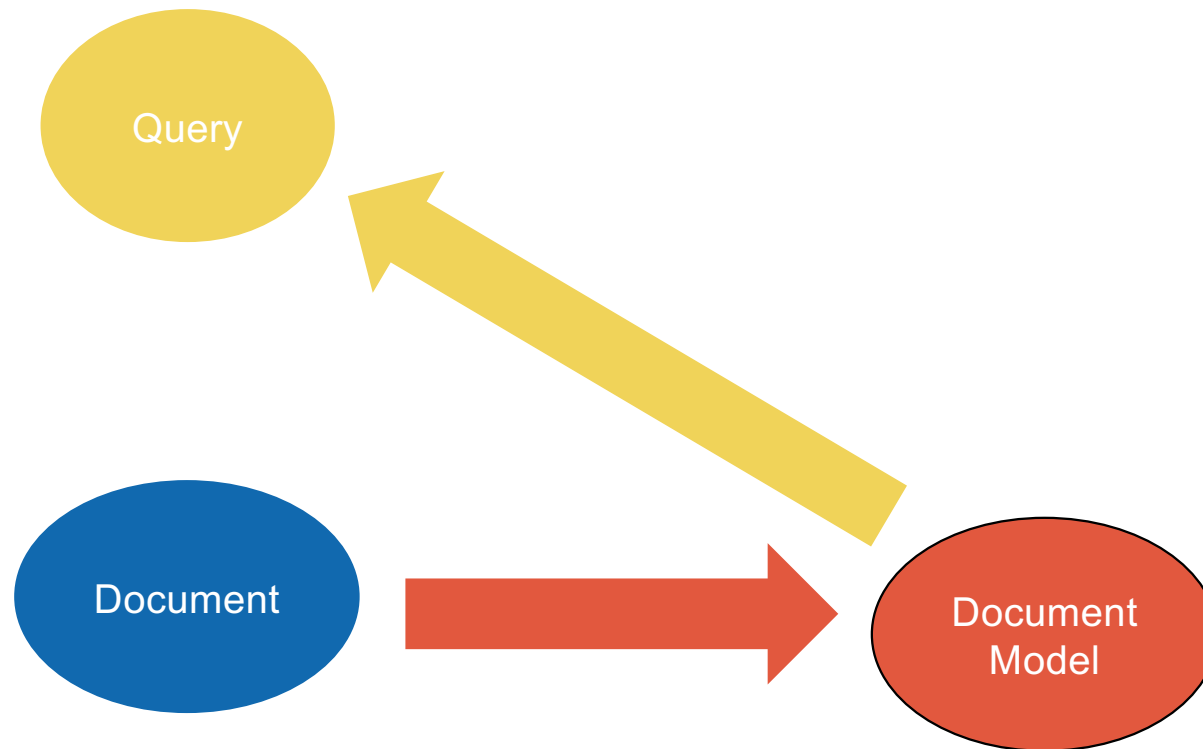
Smoothing

$$\frac{tf_{t,d}}{L_d}$$

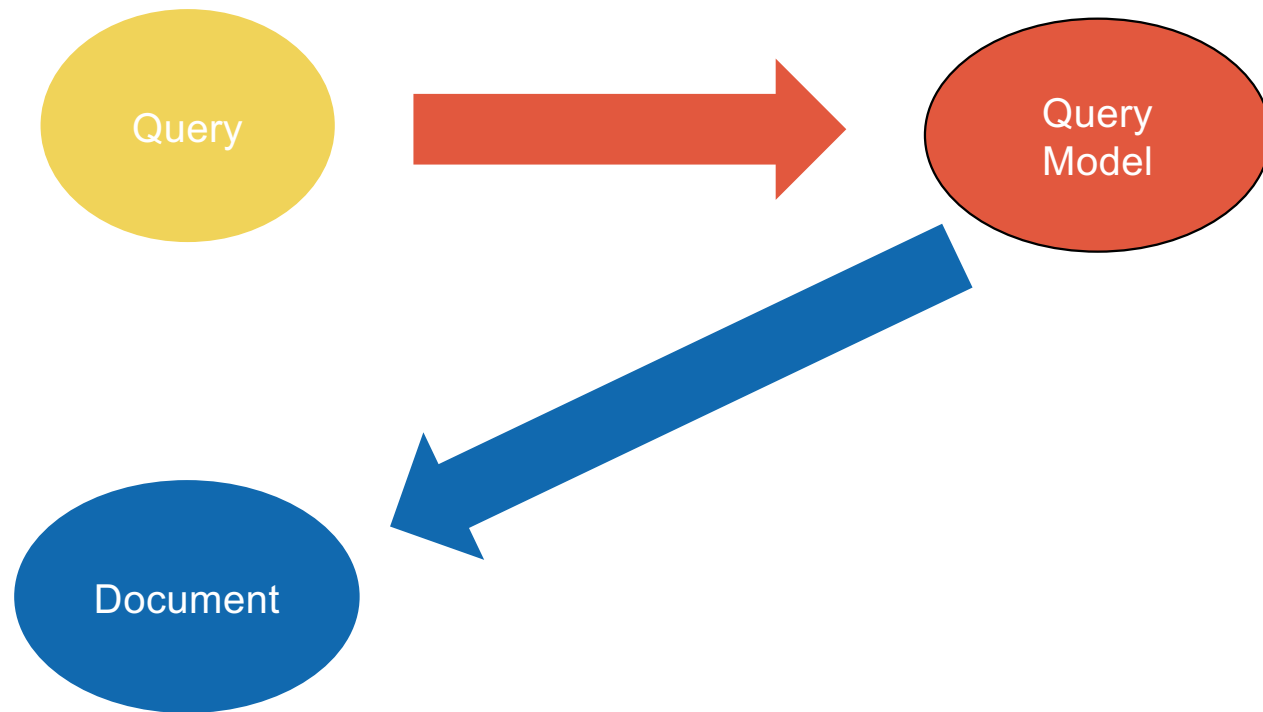


$$\lambda \frac{tf_{t,d}}{L_d} + (1 - \lambda) \frac{cf_t}{L_c}$$

Query likelihood model



Document likelihood model



Model comparison

